Pranshu Tripathi

Software Engineer

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SUMMARY

Software engineer experienced in Python, C++, FastAPI, Kubernetes, and GenAI systems; delivered 300-400x faster molecular calculations.

EXPERIENCE

Schrödinger

Senior Developer I *January 2026 - Present*

- Built Maestro Assistant's multi-agent backend and session platform, adding reusable agent tooling and mode-scoped APIs for titled, soft-deletable conversations.
- Closed a critical Maestro Assistant trust-boundary gap exposing the application database, Kubernetes API, and peer sessions; moved secrets out of images and patched high- and critical-severity CVEs.
- Eliminated a blocking ~60-second Bunsen session-pod startup interstitial through speculative pod warm-up across new, resumed, and imported sessions.

Developer II *July 2024 - December 2025*

- Engineered the "Maestro Assistant" service into a single proxy, implementing CI/CD pipelines and telemetry to support 100+ daily production requests. Received recognition from company leadership and customers like BMS, GSK.
- Implemented non-bonded interaction calculation library; a modular C++ library with caching. It reduced the calculation time by 300-400x; cut 10-min tasks to 30s.

Developer I *June 2023 - June 2024*

- Enhanced export functionality for new movie presets (e.g., View Ligand Docking) from Maestro to PyMOL, decreasing manual visualization setup time by 60% and enhancing automated workflow efficiency.

Intern - Software Development *January 2023 - May 2023*

- Created an adapter for Schrödinger's structure class that removed the m2io dependency and enabled a property-writing mechanism with 3x greater flexibility, gaining recognition from Scientific Infrastructure team.

Intern - Software Development *May 2022 - July 2022*

PROJECTS

EventHive

Deployed a ticketing platform on GKE handling 3,000+ RPS, publishing NPM packages to streamline microservice integration across 5 services

PlotCortex

A place to semantically search movies and TV shows in IMDB movies dataset

CarbonHero

Built a leaderboard-based Flutter app for GDSC Solution Challenge 2021 to track user carbon footprint, integrating Firebase for real-time updates and auth

EDUCATION

Georgia Institute of Technology
Master of Science in Computer Science
Aug 2026 - Present

- Coursework: High-Performance Computing.

IIIT Allahabad (B.Tech)

Electronics and Communication
Aug 2019 - May 2023

- **GPA: 9.16/10**
- **Batch Rank: 4**
- **Award:** Merit-cum-Means Scholarship for Academic Excellence 2020, 2021, 2022

AWARDS

- Runner-up in Schrödinger Hackathon 2024
- Appreciation from CEO, CTO for Maestro Assistant Project
- Silver Medal - Football, Badminton & Table Tennis at Schrödinger Sports Meet

SKILLS

Programming Languages

- C++, Python, Java, JavaScript, TypeScript, SQL, Shell Scripting, Terraform

Framework & Libraries

- Spring, Express.js, React, Next.js, Qt, Hibernate, PostgreSQL, SIP (Python-C++), PyQt6, BULLJS

Cloud & DevOps

- Docker, Kubernetes (K8s), GitHub Workflows, Travis CI, AWS (Elastic Beanstalk, RDS, ElastiCache), GCP, GKE, G Cloud SQL, NATs Streaming

Generative AI Technologies

- LiteLLM, Vertex AI, Hugging Face, Qdrant Vector DB, LangChain, DSPY, MCP, RAG

Additional Interests

- CNNs, LSTMs, Transformers, Graph DB, IoT Systems, Classification Models, Speech to Text recognition

VOLUNTEERING

- Coordinator - IoT Wing, IIIT Allahabad
- Senior Member - Literature Society, IIIT Allahabad

LANGUAGES

English, Hindi, Marathi, German